

FX Options Pricing for Corporate Treasury:

A Python Implementation of the Garman-Kohlhagen Framework with Greeks, Volatility Smile, and Hedging Scenario Analysis

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github.com/imransiddiqui-cfa/treasury-intelligence-framework

Abstract

Foreign exchange options are the primary instruments through which multinational corporations and financial institutions manage currency risk while preserving the ability to benefit from favorable market movements. Yet the pricing, risk measurement, and hedging analytics underlying these instruments remain poorly understood by many corporate treasury teams — a gap that creates dependence on bank counterparties who are simultaneously on the other side of the trade.

This paper provides a complete, practitioner-oriented treatment of FX options pricing using the Garman-Kohlhagen (1983) model — the FX market's adaptation of the Black-Scholes-Merton framework — implemented in Python. The presentation proceeds from first principles: the economic rationale for options versus forwards, the Garman-Kohlhagen pricing formula and its derivation from covered interest rate parity, the complete set of option Greeks with their practical interpretation for corporate risk management, the volatility smile and its implications for OTM option pricing, and a four-strategy scenario analysis for a representative GCC/Pakistan corporate importing from Japan with JPY 500 million payable in six months.

The Python implementation — published as Module 4 of the Treasury Intelligence Framework open-source library — includes the GarmanKohlhagen pricing class with all eight Greeks (delta, gamma, vega, theta, rho-domestic, rho-foreign, vanna, volga), a volatility surface builder from standard FX market quotes (ATM, 25-delta risk reversal, 25-delta butterfly), a Greek sensitivity analyzer, a corporate hedging scenario engine comparing four strategies (unhedged, forward, vanilla option, zero-cost collar), and a Monte Carlo delta hedging simulator that demonstrates why the premium equals the expected cost of

dynamic replication. Empirical validation on the USD/JPY importer scenario confirms put-call parity to within 3×10^{-7} and produces a delta hedging simulation mean P&L; of approximately zero — consistent with the Black-Scholes no-arbitrage condition.

1. Introduction: Why Corporates Need Options, Not Just Forwards

An FX forward contract eliminates currency risk completely. For a Japanese importer owing JPY 500 million in six months, a forward locks in today's rate adjusted for the interest rate differential between the two currencies — the forward rate. The importer knows exactly what they will pay in domestic currency terms, regardless of where spot trades at maturity. This certainty has value.

But certainty has a cost: the importer who locks in a forward at USD/JPY 153.91 (the six-month forward given current rate differentials) receives no benefit if the yen weakens to 170 at maturity — a rate at which buying yen spot would have cost significantly less than the forward commitment. The forward eliminates both the bad outcome (yen strengthens) and the good outcome (yen weakens). For many corporate treasury functions operating under budgeted exchange rates, this trade-off is acceptable. For those with profit-sensitive FX exposure, it is not.

"An FX option gives the corporate the right, but not the obligation, to exchange currencies at a predetermined rate. The buyer pays a premium for this asymmetry — protection against adverse moves with full participation in favorable ones. The bank sells the option and receives the premium, taking on the obligation to fulfill the contract if exercised."

Understanding what drives that premium — and how it changes as market conditions evolve — is essential for any treasury professional evaluating hedging alternatives. The Garman-Kohlhagen model provides the mathematical framework. The Greeks translate that framework into actionable risk metrics. The scenario analysis makes the trade-offs concrete.

2. Key Terminology

Term	Definition	Example (USD/JPY Importer)
Strike (K)	Exchange rate locked in by the option	K = 150: right to buy JPY at 150
Spot (S)	Current market exchange rate	S = 150 USD/JPY today

Expiry (T)	Date option ceases to exist	6 months = T = 0.5 years
Call Option	Right to buy the base currency	Right to buy USD (sell JPY)
Put Option	Right to sell the base currency	Right to sell USD (buy JPY) — what importer needs
Premium	Price paid upfront for the option right	USD ~177k per USD 1M notional
ITM	Option has intrinsic value if exercised now	Put ITM if spot falls below strike
OTM	No intrinsic value; only time value remains	Put OTM if spot rises above strike
ATM	Strike equals current spot; maximum time value	K = S = 150
Delta (Δ)	Premium sensitivity per 1-unit spot move	Put $\Delta \approx -0.314$
Implied Vol (σ)	Market's consensus forecast of future spot swings	8% for USD/JPY 6M
Forward Rate (F)	Spot adjusted for interest rate differential	F = 153.91 (USD rates > JPY)

3. The Garman-Kohlhagen Pricing Model

3.1 Theoretical Foundation

The Garman-Kohlhagen (1983) model extends the Black-Scholes-Merton framework to foreign exchange by recognizing that a foreign currency earns the foreign interest rate (r_f) continuously — analogous to a dividend yield on a stock. Under risk-neutral measure, the spot exchange rate S follows geometric Brownian motion:

$$dS = (r_d - r_f) \times S \times dt + \sigma \times S \times dW$$

where r_d is the domestic interest rate, r_f is the foreign interest rate, σ is the instantaneous volatility, and dW is a standard Brownian increment. The drift ($r_d - r_f$) ensures the model is consistent with covered interest rate parity: the expected spot rate at time T equals the forward rate $F = S \times \exp((r_d - r_f) \times T)$.

3.2 The Pricing Formula

The Garman-Kohlhagen closed-form solutions for vanilla calls and puts are:

$$\text{Call: } C = S \times e^{(-r_f \times T)} \times N(d1) - K \times e^{(-r_d \times T)} \times N(d2)$$

$$\text{Put: } P = K \times e^{(-r_d \times T)} \times N(-d2) - S \times e^{(-r_f \times T)} \times N(-d1)$$

where $N(x)$ is the cumulative standard normal distribution function, and:

$$d1 = [\ln(S/K) + (r_d - r_f + \sigma^2/2) \times T] / (\sigma \times \sqrt{T})$$

$$d2 = d1 - \sigma \times \sqrt{T}$$

Economic Interpretation of d1 and d2:

d2 is the risk-neutral probability that the option expires in the money ($N(d2)$ for a call, $N(-d2)$ for a put). $d1$ is $d2$ adjusted for the "moneyness delta" — the convexity benefit the option buyer receives from the asymmetric payoff structure. $N(d1)$ is the option's delta.

3.3 Worked Numerical Example — USD/JPY Importer

Market inputs: $S = 150$, $K = 150$ (ATM), $r_d = 5.25\%$ (USD SOFR), $r_f = 0.10\%$ (JPY policy rate), $\sigma = 8\%$, $T = 0.5$ years.

$$\text{Forward rate: } F = 150 \times \exp((0.0525 - 0.001) \times 0.5) = 153.91$$

$$d1 = [\ln(1) + (0.0525 - 0.001 + 0.0032) \times 0.5] / (0.08 \times 0.7071) = 0.4835$$

$$d2 = 0.4835 - 0.0566 = 0.4269$$

$$N(d1) = 0.6858 \mid N(d2) = 0.6652 \mid N(-d2) = 0.3348$$

$$\text{Put} = 150 \times e^{(-0.0525 \times 0.5)} \times 0.3348 - 150 \times e^{(-0.001 \times 0.5)} \times (0.6858 - 1)$$

$$\text{Put premium} = 1.7744 \text{ USD per 1 USD of notional} \Rightarrow 1.77\% \text{ of notional}$$

For JPY 500 million payable ($\approx$ USD 3.33 million notional at spot 150), the ATM put option premium is approximately USD 5,915,000, or 1.18% of the JPY notional. The 6-month forward rate of 153.91 reflects the fact that higher USD rates make yen cheaper to lock in on a forward basis — the forward is above spot, meaning the yen appears "cheaper" in forward space.

3.4 Python Implementation

```

from tif.derivatives.fx_options import GarmanKohlhagen, FXOptionInputs

gk = GarmanKohlhagen()

# ATM USD put / JPY call — importer buying JPY protection
inp = FXOptionInputs(
    spot=150.0,    strike=150.0,
    rd=0.0525,    rf=0.0010,    # USD SOFR vs JPY policy rate
    vol=0.08,     T=0.5,        # 8% implied vol, 6-month expiry
    option_type="put",
    notional=3_333_333,        # USD notional (JPY 500M / 150)
)

result = gk.price(inp)
print(result.summary())

# Output (selected):
# PREMIUM      1.7744 per unit
#              5,914,725 total (notional: 3,333,333)
# Delta:  -0.3142   Gamma:  +0.04181
# Vega:   +0.3763   Theta:  -0.001342 per day
# Prob ITM: 33.5%

```

4. The Greeks — Complete Reference

4.1 Delta (Δ) — Spot Sensitivity

Delta is the rate of change of the option premium with respect to the spot rate. It represents the hedge ratio: the number of units of the underlying currency the option writer must hold to be instantaneously delta-neutral.

Call delta: $\Delta = e^{(-rf \times T)} \times N(d1)$ [ranges 0 to +1]

Put delta: $\Delta = e^{(-rf \times T)} \times (N(d1) - 1)$ [ranges -1 to 0]

For our ATM put, delta = -0.3142. This means: (a) the option premium falls by 0.3142 for every 1-unit rise in spot (yen weakening — bad for the option); (b) the bank that sold the put must short USD 1,047,397 in spot to hedge; (c) the option has approximately a 33% probability of expiry in the money (delta $\approx$ probability ITM for OTM options, though not exactly for ATM). Delta of -0.50 would indicate an exactly ATM option on a forward basis.

4.2 Gamma (Γ) — Delta Curvature

Gamma = $[e^{(-rf \times T)} \times N'(d1)] / [S \times \sigma \times \sqrt{T}]$

Gamma measures how rapidly delta itself changes as spot moves. Gamma is always positive for both calls and puts (long option positions benefit from large moves in either direction). ATM options near expiry

have the highest gamma — delta can swing from near zero to near one over a small spot range, requiring the bank to rebalance its delta hedge aggressively. This rebalancing cost is borne by the option buyer through the premium. For our 6-month ATM put, gamma = 0.0418 — meaning for a 1-JPY move in spot, the delta changes by 0.0418 units, requiring the bank to rebalance its hedge position.

4.3 Vega (v) — Volatility Sensitivity

$$\text{Vega} = S \times e^{(-rf \times T)} \times N'(d1) \times \text{sqrt}(T) \text{ [per 1\% vol change]}$$

Vega measures premium sensitivity to implied volatility. Our put has vega = +0.3763 (per 1%), meaning: if USD/JPY implied vol rises from 8% to 9%, the option gains USD 1,254,263 in value — even if spot has not moved. This is the primary way option books gain or lose value over time, independent of the underlying exchange rate. The practical implication for corporates: buying options when implied volatility is low is significantly cheaper than buying during crisis episodes when volatility spikes. The 2022 yen depreciation saw USD/JPY implied vol exceed 12%, making 6-month puts approximately 50% more expensive than at 8% vol.

4.4 Theta (Θ) — Time Decay

Theta measures the daily erosion of option value from the passage of time, holding spot and volatility constant. For long option positions, theta is always negative — time works against the buyer. For our put, theta = -0.001342 per day, implying a daily loss of USD 4,473 on the total position. This accelerates non-linearly as expiry approaches: an ATM option that loses 0.1% of value daily at six months may lose 0.5% daily in the final two weeks.

Theta vs. Gamma Trade-off:

Long options pay theta daily but receive gamma: large spot moves cause the option to gain more than theta extrapolation would predict, due to delta rebalancing. Short options (sold by the bank) earn theta but pay gamma. The at-the-money option sits precisely at the point of maximum gamma and maximum theta — highest sensitivity to both spot moves and time.

4.5 Rho (ρ) — Interest Rate Sensitivity

$$\text{Rho}_d = K \times T \times e^{(-rd \times T)} \times N(d2) \times 0.01 \text{ [domestic, per 1\% rate move]}$$

$$\text{Rho}_f = -T \times S \times e^{(-rf \times T)} \times N(d1) \times 0.01 \text{ [foreign, per 1\% rate move]}$$

Rho measures sensitivity to interest rates. For our put, rho_d = -0.2445 (domestic USD rate) and rho_f = +0.2357 (foreign JPY rate). Rising USD rates increase the forward rate (making the spot put less valuable) while rising JPY rates decrease the forward rate (making the spot put more valuable). Rho is less important than delta, gamma, or vega for short-dated options but becomes material for maturities beyond one year.

4.6 Vanna and Volga — Second-Order Cross-Greeks

Vanna ($d^2V/dS d\sigma = -0.8931$ per unit) measures how delta changes as volatility changes. It is important for hedging in environments where spot and volatility move together — a common pattern in currency crises where sharp spot depreciation drives simultaneous vol spikes. Volga ($d^2V/d\sigma^2$, also called vomma) measures the convexity of the option price with respect to volatility. Positive volga means long option positions gain disproportionately from large vol moves — the same asymmetry in vol space that gamma provides in spot space. Banks use vanna and volga to price exotic options and construct the volatility smile.

Greek	Formula Basis	Our ATM Put Value	Corporate Risk Interpretation
Delta (Δ)	dV/dS	-0.3142	Premium falls 0.314 per 1-JPY spot rise; ~31% hedge ratio
Gamma (Γ)	d^2V/dS^2	+0.0418	Delta changes 0.042 per 1-JPY move; protection accelerates ITM
Vega (v)	$dV/d\sigma$ (per 1%)	+0.3763	USD +1.25M gain if vol rises 1pp; buy options when vol is low
Theta (Θ)	dV/dT (per day)	-0.00134	USD -4,473 per day from time decay; accelerates near expiry
Rho_d (ρ_d)	dV/drd (per 1%)	-0.2445	USD rates rising 1% reduces put value by USD 815k
Rho_f (ρ_f)	dV/dr_f (per 1%)	+0.2357	JPY rates rising 1% increases put value by USD 786k
Vanna	$d^2V/dS d\sigma$	-0.8931	Delta becomes more negative when vol rises (crisis correlation)
Volga	$d^2V/d\sigma^2$	+0.0097	Convex in vol; option gains extra value from vol spikes

5. The Volatility Smile and OTM Option Pricing

5.1 Why the Smile Exists

The Black-Scholes model assumes constant volatility across all strikes — a theoretical convenience that does not hold in practice. In FX markets, implied volatility varies systematically with the strike, forming a curve known as the "volatility smile" (symmetric in some currency pairs) or "skew" (asymmetric where one tail is consistently more expensive).

For USD/JPY, the smile exhibits negative skew: out-of-the-money yen calls (USD puts — the protection the importer wants) trade at higher implied volatility than ATM options. This reflects the asymmetric risk perception in the market: USD/JPY is prone to sharp, rapid yen strengthening episodes (as in 2022-2023 yen carry trade unwinding) that drive tail demand for deep OTM yen calls.

5.2 Market Conventions: ATM, RR25, BF25

The FX options market quotes the volatility surface through three standardized instruments per tenor, rather than quoting volatility at specific strikes:

Quote	Full Name	Definition	Interpretation
ATM	At-the-money straddle vol	Vol of the 50-delta straddle	Baseline vol level for the tenor
RR25	25-delta risk reversal	$\text{Vol}(\text{call } 25\Delta) - \text{Vol}(\text{put } 25\Delta)$	Skew direction: negative = put premium (yen call demand)
BF25	25-delta butterfly	$[\text{Vol}(\text{call } 25\Delta) + \text{Vol}(\text{put } 25\Delta)]/2 - \text{ATM}$	Smile curvature: positive = wings above ATM

From these three quotes, the full smile is recovered: $\text{Vol}(\text{call } 25\Delta) = \text{ATM} + \text{BF25} + 0.5 \times \text{RR25}$, and $\text{Vol}(\text{put } 25\Delta) = \text{ATM} + \text{BF25} - 0.5 \times \text{RR25}$. For USD/JPY at a typical $\text{RR25} = -1.5\%$ and $\text{BF25} = +0.8\%$, the 25-delta put trades at $\text{vol} = 8.0 + 0.8 - 0.75 = 8.05\%$, while the 25-delta call trades at $8.0 + 0.8 + 0.75 = 9.55\%$.

5.3 Impact on Corporate Hedging Cost

The smile has direct practical consequences for the corporate choosing between different strike levels. An OTM put offering protection only if the yen strengthens significantly (say, strike at 143 versus current spot 150) is cheap in absolute premium terms — but trades at a higher implied volatility than ATM, partially offsetting the lower intrinsic probability of exercise:

Strike	Moneyness	Impl. Vol	Premium (USD)	% of JPY Notional	Protection Level
135	Deep OTM (10Δ)	8.45%	~USD 925	0.19%	Only if JPY appreciates 10%+
143	OTM (25Δ)	8.45%	~USD 10k	2.0%	If JPY appreciates ~5%
150	ATM (50Δ)	8.00%	~USD 5.9M	1.18%	Any JPY appreciation vs 150
148	Slight OTM	8.15%	~USD 3.8M	0.76%	JPY appreciates beyond 148

The deep OTM put at strike 135 appears attractively cheap at 0.19% of notional. However, if the yen strengthens from 150 to 140 — a significant 6.7% move — the importer with the 135-strike put remains completely unprotected. The ATM option at 1.18% of notional provides protection from the first pip of yen strengthening. The choice depends on the corporate's risk tolerance and whether they are managing to a maximum loss budget or an average cost target.

6. Corporate Hedging Strategy Comparison

6.1 Four Strategies for the Japanese Importer

A GCC or Pakistani corporate owing JPY 500 million in six months faces four primary hedging choices. The following comparison uses the USD/JPY market inputs established above (spot 150, 6M forward

153.91, ATM vol 8%):

Strategy	Mechanism	Upfront Cost	Protection	Upside Participation	Best If...
Unhedged	No action; buy JPY at spot at maturity	Zero	None	Full	Yen weakens significantly
Forward Hedge	Lock in rate at forward = 153.91	Zero (margin may apply)	Full below 153.91	None	Rate certainty is paramount
Vanilla Put	Buy USD put / JPY call at K=150	USD 5.9M (1.18% of JPY notional)	Full below 150	Full above 150	Want full protection + upside
Zero-Cost Collar	Buy put at K=150, sell call at K≈152	Zero net premium	Full below 150	Capped at 152	Want protection without premium cost

6.2 Scenario Analysis: USD Cost of JPY 500M Across Spot Scenarios

Spot at Expiry	Unhedged USD	Forward USD	Option USD	Collar USD	Option P&L;
130 (JPY +13%)	3,846,154	3,248,596	3,333,333+prem	3,333,333	+USD 60.8M
140 (JPY +6%)	3,571,429	3,248,596	3,333,333+prem	3,333,333	+USD 27.4M
150 (unchanged)	3,333,333	3,248,596	3,333,333+prem	3,333,333	-USD 5.9M
155 (JPY -3%)	3,225,806	3,248,596	3,225,806+prem	3,225,806	-USD 5.9M
165 (JPY -9%)	3,030,303	3,248,596	3,030,303+prem	3,162,955	-USD 5.9M

Key Reading of the Scenario Table:

The "Option USD" column shows the cost of buying JPY at the better of spot or strike, plus the premium. At spot=130 the option allows buying JPY at 150 (protecting against a 13% yen appreciation), saving ~USD 513k versus spot while the unhedged position would cost USD 3.85M. The option P&L of +USD 60.8M at spot=130 represents the option payoff net of premium — the full economic gain from protection. At spot=155 (yen weaker), the option expires worthless and the corporate benefits from buying yen cheaper in spot, net of premium cost.

7. Why the Premium Equals What It Does: Delta Hedging

The Black-Scholes premium is not an arbitrary price set by the bank — it is the theoretical cost of replicating the option's payoff through continuous dynamic hedging of the delta position. The bank that sells a put option immediately buys approximately 0.314 units of USD for every unit of notional to delta-hedge. As spot moves, the delta changes (due to gamma) and the bank must continuously rebalance. The cumulative cost of this rebalancing, under the risk-neutral measure, equals exactly the option

premium.

This is demonstrated rigorously through the Monte Carlo DeltaHedgingSimulator in the Treasury Intelligence Framework library. Simulating 5,000 geometric Brownian motion paths and computing the bank's hedging P&L; on each path produces a mean P&L; of approximately zero — confirming that the 1.7744 premium is precisely sufficient to cover the expected dynamic hedging cost. The distribution of hedging P&L; has a standard deviation of approximately 0.02 per unit, reflecting residual gamma risk from discrete (daily) rather than continuous rebalancing.

```
from tif.derivatives.fx_options import DeltaHedgingSimulator, FXOptionInputs

sim = DeltaHedgingSimulator(n_steps=126, n_paths=5000)
inp = FXOptionInputs(spot=150, strike=150, rd=0.0525,
                    rf=0.001, vol=0.08, T=0.5,
                    option_type="put", notional=1.0)

result = sim.simulate(inp)
print(f"Theoretical premium: {result['theoretical_premium']:.6f}")
print(f"Mean hedging P&L: {result['mean_pnl']:.6f} (~0 confirms pricing)")
print(f"Std hedging P&L: {result['std_pnl']:.6f} (gamma risk)")
print(f"5th percentile P&L: {result['pnl_percentile_5']:.6f}")
print(f"95th percentile P&L: {result['pnl_percentile_95']:.6f}")
```

8. Governance Implications for Corporate Treasury

8.1 Independence from Bank Counterparty Advice

A corporate treasury that understands option pricing can independently verify whether the premium quoted by the bank is consistent with prevailing market implied volatility levels. The GarmanKohlhagen.implied_vol() method allows back-solving from a quoted premium to the embedded implied volatility — revealing whether the bank is pricing at market vol or adding a spread. For a USD 3.33M notional 6-month ATM put, a 1 vol-point premium spread (quoting at 9% rather than 8%) increases the premium by approximately USD 125,000 — the vega of the position. This is invisible to a corporate that cannot independently price options.

8.2 Hedge Effectiveness Under IFRS 9 and ASC 815

FX option hedges qualify for hedge accounting treatment under both IFRS 9 and ASC 815 when properly documented and tested for effectiveness. The critical governance requirement is that the option's intrinsic value — not its time value — serves as the designated hedging instrument for most FX cash flow hedges. Time value changes (theta decay) are recorded separately in OCI and released to earnings over the hedge period. The automated effectiveness testing module in the Treasury Intelligence Framework (Module 2) handles this designation and testing workflow automatically, eliminating the primary source of ASC 815

misclassification errors that generate unnecessary earnings volatility.

8.3 Practical Decision Framework

Decision Factor	Consider Forward	Consider Vanilla Option	Consider Collar
Premium budget	No premium concern	Premium affordable in budget	Must avoid premium outlay
Rate uncertainty	High (need certainty)	Moderate (want protection + upside)	Moderate
Accounting preference	Simple (no time value split)	IFRS 9/ASC 815 documented hedge	Same as option
Vol environment	Irrelevant	Buy when vol is low (<8%)	Buy when vol is low
Tenor	All tenors	Up to 12 months most liquid	Up to 12 months most liquid
Payment certainty	Confirmed invoices only	Forecast exposure acceptable	Forecast acceptable

9. Conclusion

FX options provide corporate treasury teams with an asymmetric risk management tool that forwards cannot replicate: protection against adverse exchange rate moves with full participation in favorable ones. The premium paid for this asymmetry is not arbitrary — it is the mathematically precise cost of dynamically replicating the option's payoff, as demonstrated both by the Garman-Kohlhagen closed-form solution and the Monte Carlo delta hedging simulation presented in this paper.

The complete Python implementation — published as Module 4 of the open-source Treasury Intelligence Framework — gives corporate treasury professionals and bank ALM practitioners the tools to independently price options, monitor Greeks in real time, construct volatility smiles from market quotes, and compare hedging alternatives across rate scenarios. The ability to independently price a derivative that a bank counterparty is quoting is the foundation of informed hedging governance.

The Japanese importer case study — JPY 500 million payable in six months, USD/JPY spot 150, 8% implied volatility — demonstrates that an ATM put option costing 1.18% of notional provides full downside protection while preserving the full benefit of yen weakening. The zero-cost collar eliminates the premium but caps the upside benefit at the call strike. Neither is universally superior: the optimal choice depends on the corporate's budget, accounting framework, and view on yen direction. What matters is that the choice is made with full analytical transparency — not based on a bank's recommendation alone.

About the Author

Imran Siddiqui, CFA, has over 20 years of executive treasury leadership across HSBC Bank Oman, National Bank of Oman, Sohar International Bank, and Bahri (National Shipping Company of Saudi Arabia). He has designed and governed derivative hedging programs covering interest rate swaps, cross-currency swaps, FX forwards, FX options, par forwards, and structured collars. He has negotiated ISDA Master Agreements and Credit Support Annexures with international banking counterparties, and managed investment portfolios exceeding USD 3 billion in notional value. His Computer Science undergraduate education enables direct Python development of the analytical systems described in this paper. He holds a B.Sc. Computer Science (Hamdard University), MBA (SZABIST), and the CFA charter.

Disclosure

The views expressed are those of the author in a personal professional capacity. Python implementations are illustrative and do not constitute production-ready or audited software. All numerical examples use indicative market data. This article is submitted for practitioner publication consideration (GARP Risk Intelligence / AFP Exchange / Journal of Applied Corporate Finance) and may be submitted in support of an EB-2 National Interest Waiver immigration petition.

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